Unit 9, Lesson 3: Subtracting Linear Expressions with Rational Coefficients

Benchmark

MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.

Learning Target

☐ I can subtract linear expressions with rational coefficients.

9.3.1 Warm-Up: Bell Work

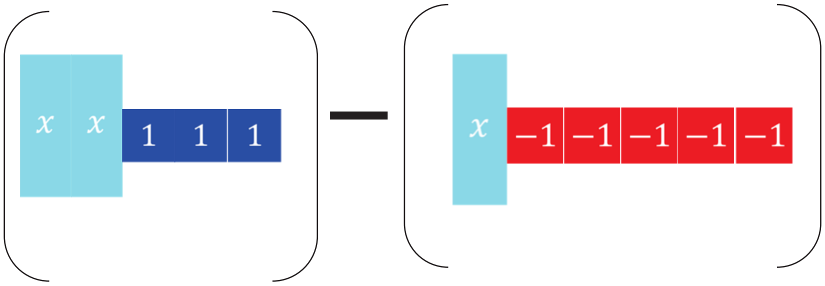
Combine like terms.

1. $(3y + 4.6) + (-5y - 3.25)$

2. $\left(6\frac{3}{4}x - 7\right) + \left(-\frac{5}{2}x + 7\right)$

9.3.2 Exploration: Subtracting Expressions with Algebra Tiles

1. Algebra tile representations are given below. Use the algebra tiles to find the difference.

On the lines, write the algebraic expression for each representation.	 ________ – ________
Rearrange the tiles to show subtraction of like tiles together. Draw the representation.	
If necessary, create any zero pairs needed for subtraction.	
Find the difference by subtracting (taking away) like tiles. Write the algebraic representation of the difference.	

9.3.3 Collaboration: Subtracting Expressions

1. Noah, Lin, Jada, and Andre are rewriting the expression $(x + 8) - (4 - 9x)$ as an equivalent expression with fewer terms. Each student described their work.

   Discuss with your partner which students made an error. Describe each student's error and correct the error.

   a. Noah says, "I dropped the parentheses, combined like terms, and ended up with $-8x + 4$."

$$(x + 8) - (4 - 9x)$$
$$x + 8 - 4 - 9x$$
$$-8x + 4$$

   b. Lin says, "I simplified inside the parentheses and ended up with $13x$."

$$(x + 8) - (4 - 9x)$$
$$(8x) - (-5x)$$
$$8x + 5x$$
$$13x$$

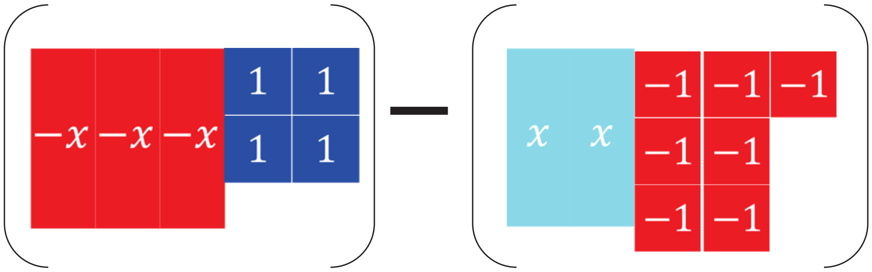
   c. Jada says, "I know that subtracting is adding the opposite, and ended up with $4 + 10x$."

$$(x + 8) - (4 - 9x)$$
$$(x + 8) + (-4 + 9x)$$
$$x + 8 - 4 + 9x$$
$$4 + 10x$$

   d. Andre says, "I used the distributive property since there is an invisible (-1) in front of the second parentheses to distribute, but I ended up with $10x + 4$."

$$(x + 8) - (4 - 9x)$$
$$(x + 8) - 1(4 - 9x)$$
$$x + 8 - 4 + 9x$$
$$10x + 4$$

2. When subtracting expressions, use the distributive property to add the opposite. With the algebra tile representations below, model finding the difference by adding the opposite.

On the lines, write the algebraic expression for each representation.	 __________ – __________
Rewrite subtracting to adding the opposite by changing (or flipping) the sign of the 2nd expression. Draw the representation.	
Combine any zero pairs to find the sum. Write the sum.	

9.3.4 Guided Instruction: Subtracting Linear Expressions

1. To subtract linear expressions,

   - add the _______________, or additive inverse of each term in the expression.

   OR

   - distribute a ___________ to each term in the expression.

2. Complete the table to show both methods.

Additive Inverse Method	Applying the Distributive Property
$(4x + 5) - (-2x + 6)$	$(4x + 5) - (-2x + 6)$

3. When adding and subtracting, parentheses are used to group expressions. Why is the use of parentheses especially important when subtracting?

9.3.5 Guided Practice: Subtracting Linear Expressions with Rational Coefficients

1. Find the difference of each expression.

a. $(3y + 3.5) - (8.2y - 6)$	b. $\left(m - \frac{3}{8}\right) - \left(3\frac{1}{4} - m\right)$
c. $\left(-2 + \frac{9}{2}y\right) - \left(5.3y - 5\frac{1}{2}\right)$	d. $(-7.5t - 3) - \left(7 + 3\frac{1}{8}t\right)$

2. What is the result when $-3x + 4.48$ is subtracted from $-2.2x + 3$?

a. Write the algebraic expression.

b. Rewrite the expression without parentheses.

c. Simplify the expression by rewriting it using the fewest terms possible.

9.3.6 Your Turn!

1. Find the difference of each expression in simplest form.

a. $(-2h + 8) - (h + 10)$	b. $\left(\frac{7}{2}x + \frac{4}{3}\right) - \left(3 - \frac{9}{8}x\right)$
c. $(-4.4y - 9) - (5 - 2.1y)$	d. $\left(2\frac{2}{3}n + 1\right) - \left(-\frac{1}{4}n - 6\right)$

2. What is the result when $\frac{5}{8}x + \frac{4}{3}$ is subtracted from $\frac{5}{4}x - \frac{31}{6}$?

9.3.7 Wrap-Up: Ticket Out the Door

Write the following expressions using the fewest possible terms.

1. $\left(5x - 13\frac{1}{4}\right) - \left(4\frac{1}{2} - 5x\right)$

2. $(0.7 - 2.4x) + \left(-5.32 + 4\frac{1}{10}x\right)$